

domain. With the 6 GB of storage space its tiny mobile hard drive carries, the Jukebox is definitely an offering for the hardcore collector who likes to lug around his entire arsenal with him. Although most such enthusiasts have a larger arsenal than that, the storage is still quite sufficient. The real bother is transferring so much of content to the Jukebox at miserable speeds, averaging at 500 KBps. At this speed it would take you more than 20-25 minutes to transfer a CDs worth of content at peak speeds.

The second in reckoning was the eXpanium, the CD MP3 player from Philips. Just pop in an MP3 CD and out comes a stream of audio. And you can carry as many CDs as you want, swap them whenever you wish. No audio transfer software, no cables, no messy and slow-speed downloads to the player.

Then come the whole bunch of solid states. These MP3 players have no moving parts and store your songs on Flash Memory which makes them very robust and compact and very expensive at the same time as compared to other mediums. The Nomad II MG was the one that came out tops in this category, with 64 MB of built-in memory and a USB connector that offers more than 350-400 Kbps of data upload speeds, which is around twice as fast as the fastest parallel port based player we received for testing. The Nomad II MG took around 1:04 minutes to complete our file transfer test, which took around 1:59 minutes on the Rio 300 (32 MB), whereas the old Nomad (32 MB) and the Philips Rush (32 MB) took more than eight minutes to finish this test. The D-Link DMP100 was better than both the Nomad and the Rush, taking around 5 minutes to finish the test.

Physical parameters

The size of the device is an important factor when it comes to handling, as a larger device would obviously be more troublesome to carry around. Though the player might have all the controls to manipulate the audio the way you want, if they aren't in the right place, operating the player becomes a hassle. A clear and visible display adds to the looks and usability of the device, as you should be able to see what song is playing. A poor build quality means a shorter lifespan for your device. MP3 players aren't cheap and these devices need to be sturdy, as you would want to take your player along when you trek, travel or jog. Besides, the MP3 player is your personal accessory so it has got to look good!

MP3 Everywhere!

A simple portable MP3 player was just the beginning. The ability to pack in an MP3 player in a package as small as a matchbox has fuelled the growth of devices that support MP3 playback as their secondary function. Devices ranging from watches, mobile phones, digital cameras to PDAs and car navigation systems are packing in the ability to act as an MP3 player, but are they really worth it?

Most such devices come with less

memory than what is optimal for storing decent playtimes of music and, as this memory is shared with other things such as photographs or GPS data, it makes it even less usable. Above that, these devices generally cost about as much as a new MP3 player and a standard similar mobile phone or digital camera, and aren't too feature-filled either. These devices are yet to become mature and usable, but provide a nice peek into the future of integrated audio players.

The **Philips Rush** was an all out winner in this category; so much so that we ended up giving it full marks on all the parameters in this particular test category. A large and clear display right in the centre, well designed and accessible controls that are well labelled and easy to understand and a compact and sleek shape. The only one that came anywhere close to the Rush was the Nomad II MG. This sleek and sturdy player has all its pieces set right



Casio WMP-IV: A show-off tool rather than a utility device

except the rubber controls which are placed on the side of the player, making them slightly harder to use as compared to a device that has buttons in front. It comes with a small remote control that allows you to control simple functions such as next song, pause, play, etc and fits in between the player and the headphones. You can use it as a tiepin or a collar controller. This eliminates the need to grope for the buttons while the player is in a carry pouch or in your pocket.

The Nomad looks just as good as the Nomad II MG, but has a very small display

that can be quite difficult to read in low light conditions. Just like the Nomad II MG, it has all its controls on its sides, but much smaller hard plastic buttons that are a pain to operate. The D-Link has well placed controls, good looks and an effective display but feels a little fragile as compared to other players. The Casio WMP-1V is small and funky but unrealistic and highly uncomfortable. It has really small and downright painful controls.

Both the Nomad Jukebox and the Philips eXpanium were just about as big as most Discmans. This much space allows for a lot of nice controls and display configurations, but the Nomad Jukebox was the only one that used it efficiently. The eXpanium has a small, one-line display that doesn't even display the name of the song playing, though the controls were just fine.

Usability

The easiest to use device would be one that is easily accessible with one-touch buttons to perform the most frequent tasks, and is well labelled. Menus are nice, but only when going into detailed levels and more complex options. Ease of use is a combination of many things when it comes to an MP3 player: easy readability of the display information, clear instructions outside the display area and the ergonomics of the controls. Display information is as necessary as the controls, as you generally need to have display for song names/numbers, playtime and extra functions such as repeat, equalizer, etc. Icons that represent a certain feature being active or inactive are also most welcome.

The **Nomad Jukebox** was a clear winner in this area, with its large and detailed display, easy-to-use and well-labelled controls and an easy interface for browsing